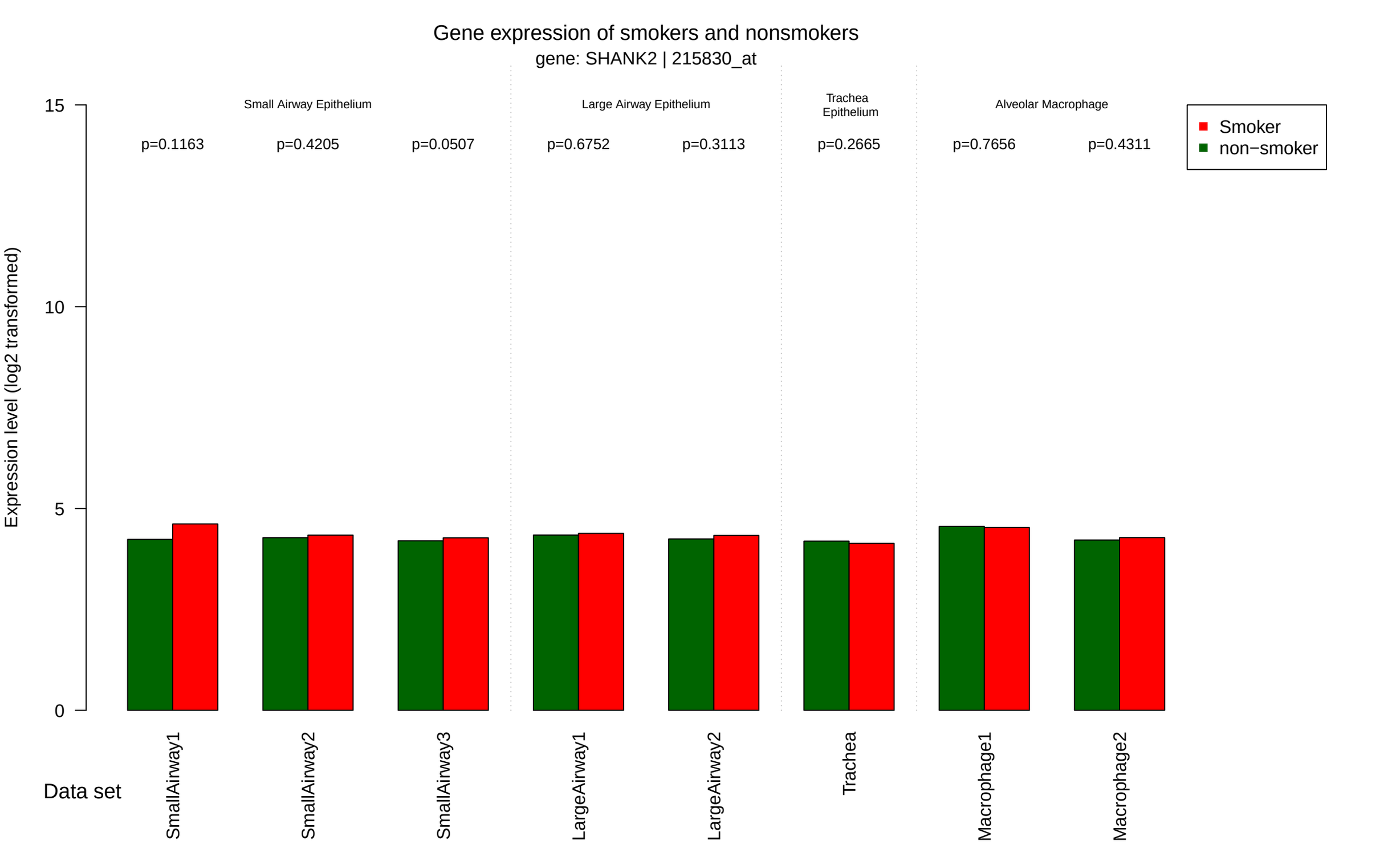
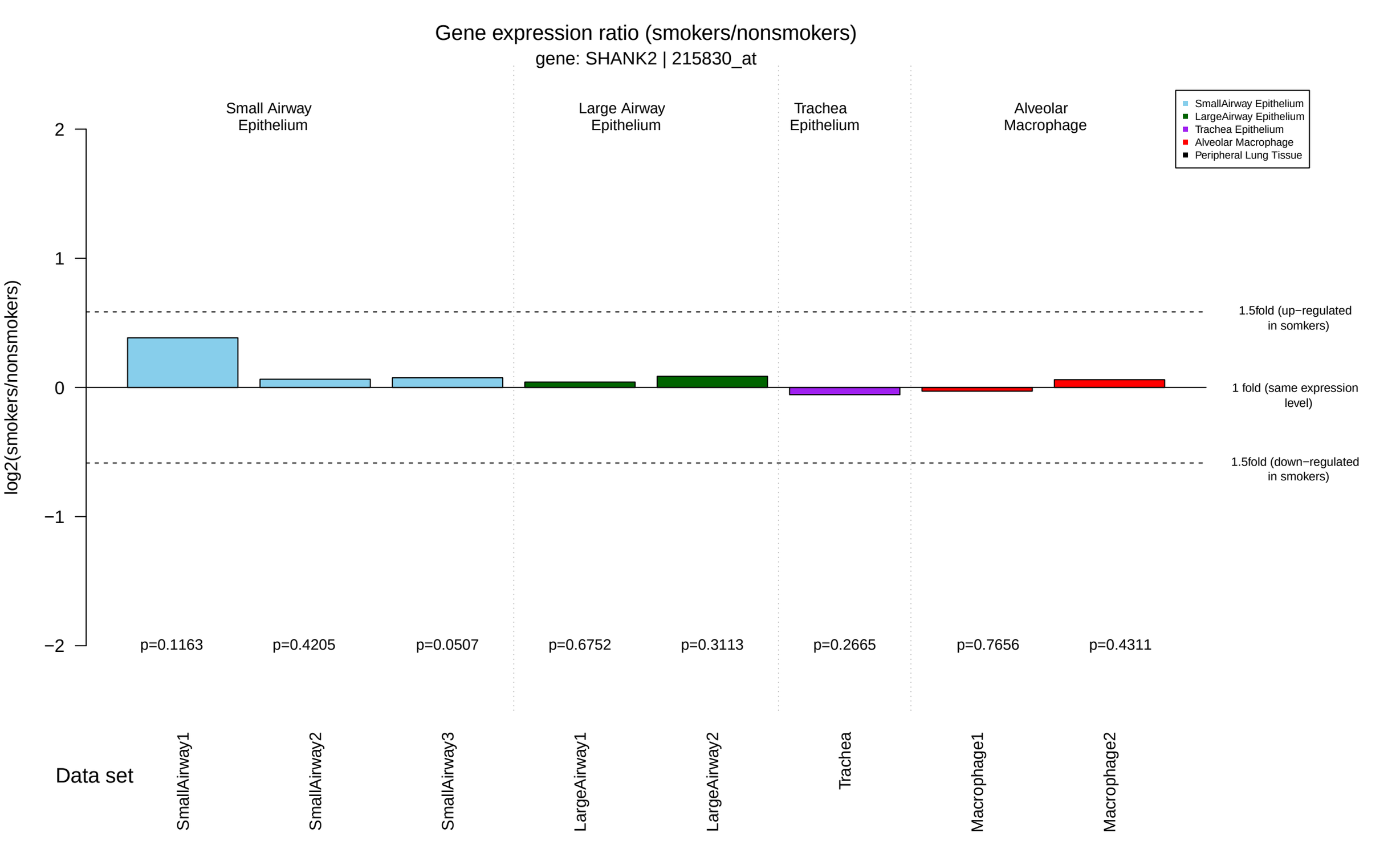
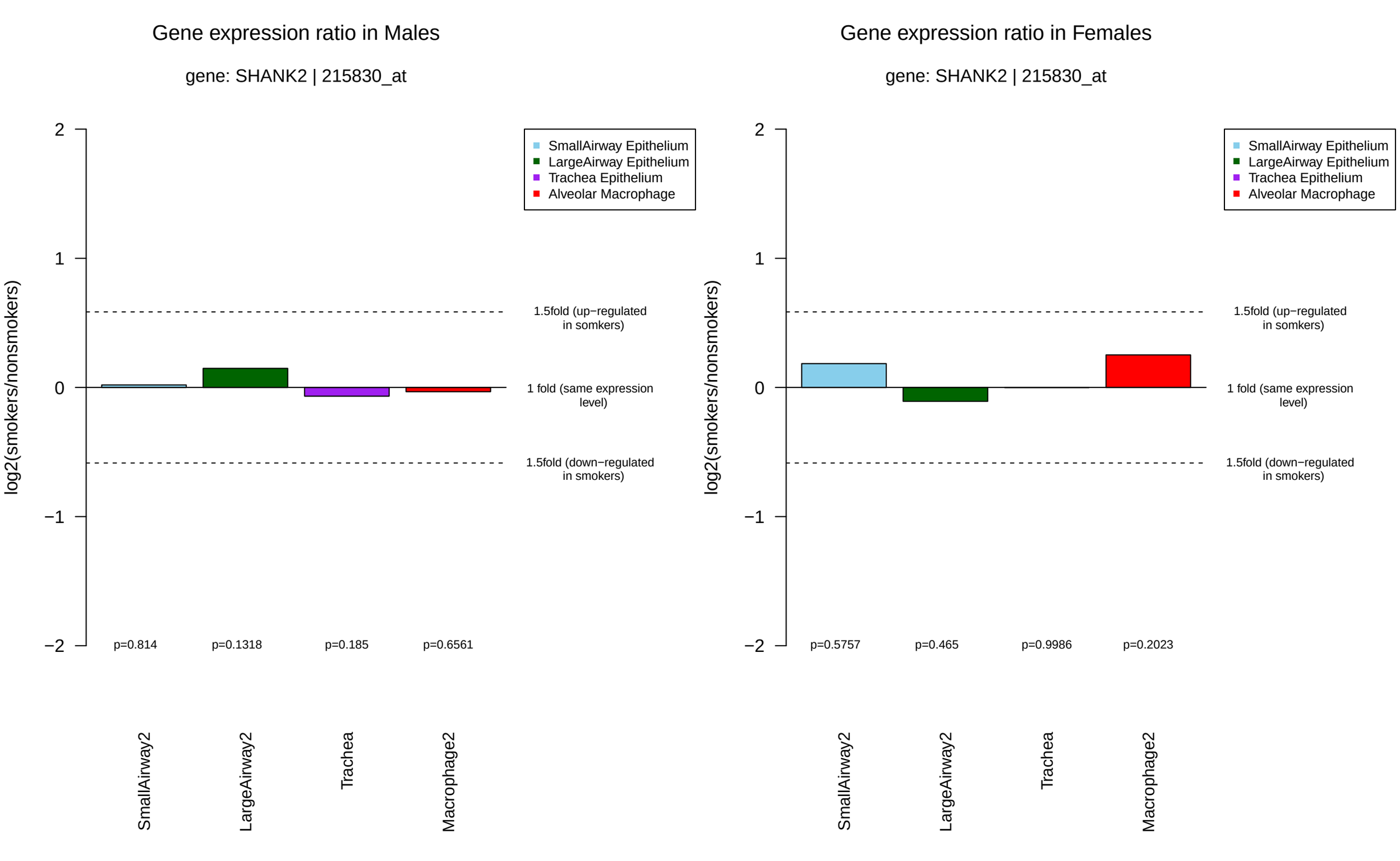
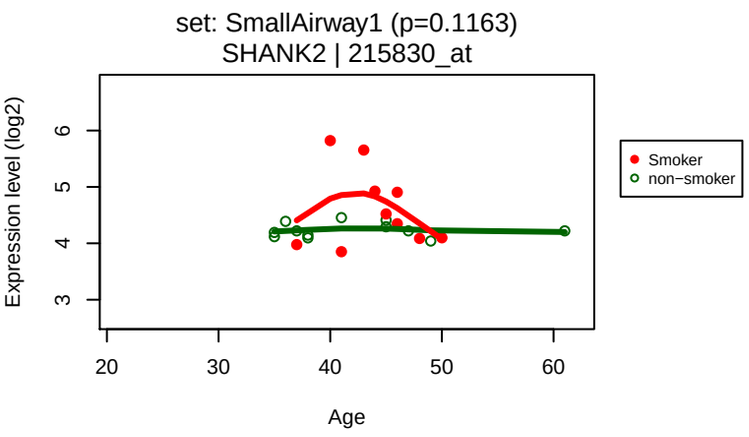
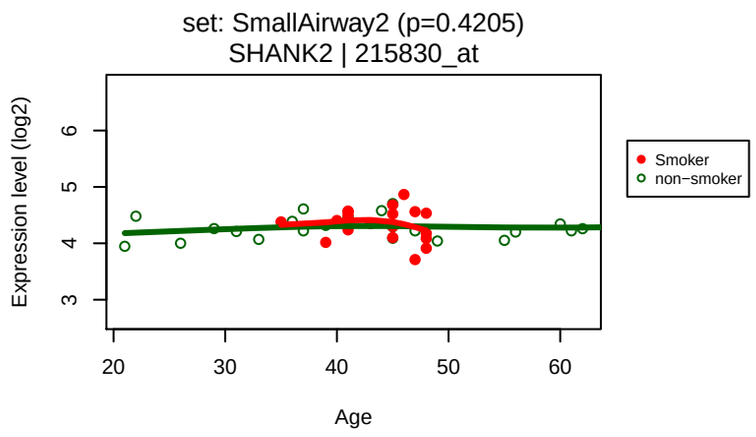
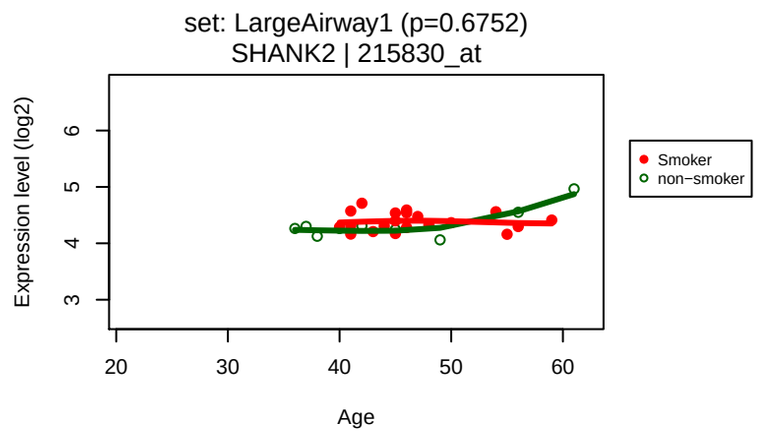

Gene expression of smokers and nonsmokers
gene: SHANK2 | 215830_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1163
p=0.4205
p=0.0507
p=0.6752
p=0.3113
p=0.2665
p=0.7656
p=0.4311
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

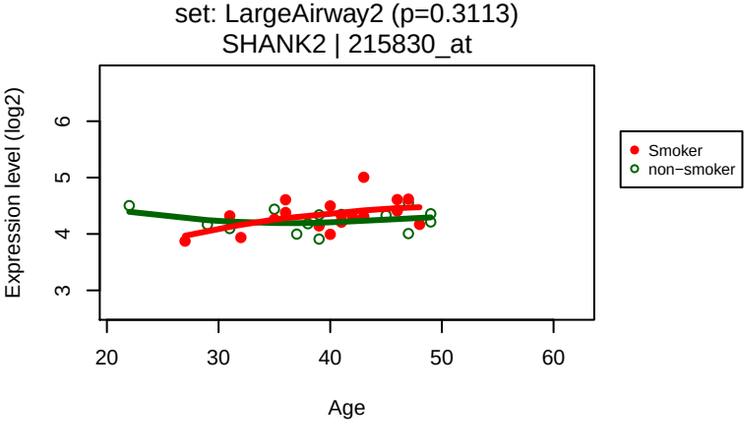

Gene expression ratio (smokers/nonsmokers)
gene: SHANK2 | 215830_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1163
p=0.4205
p=0.0507
p=0.6752
p=0.3113
p=0.2665
p=0.7656
p=0.4311
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

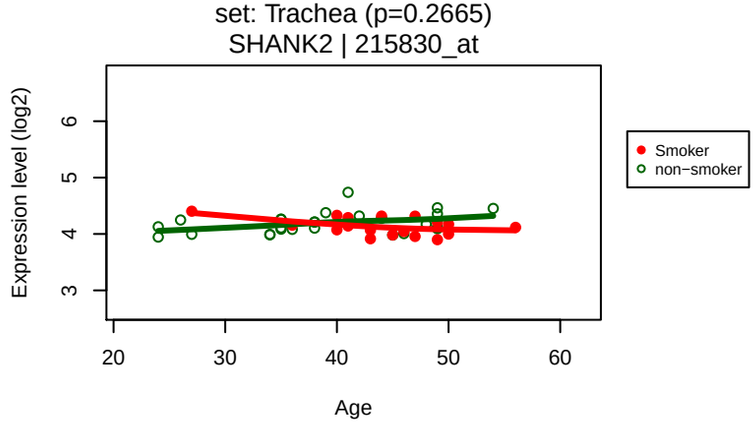

Gene expression ratio in Males
gene: SHANK2 | 215830_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.814
p=0.1318
p=0.185
p=0.6561
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SHANK2 | 215830_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5757
p=0.465
p=0.9986
p=0.2023
SmallAirway2
LargeAirway2
Trachea
Macrophage2

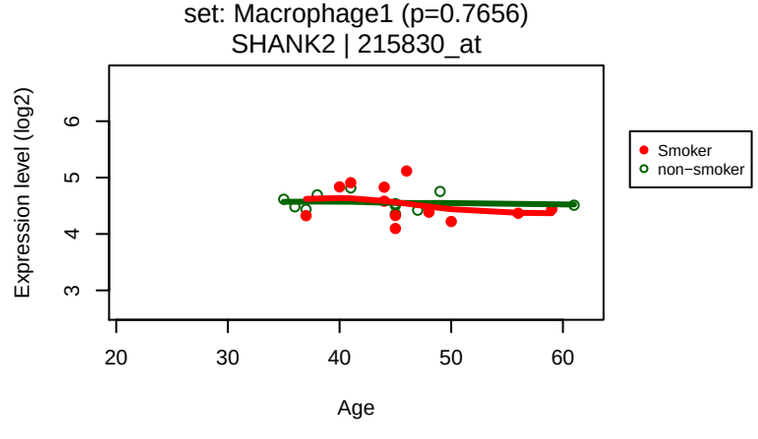

set: SmallAirway1 (p=0.1163)
SHANK2 | 215830_at
Expression level (log2)
Age
Smoker
non−smoker

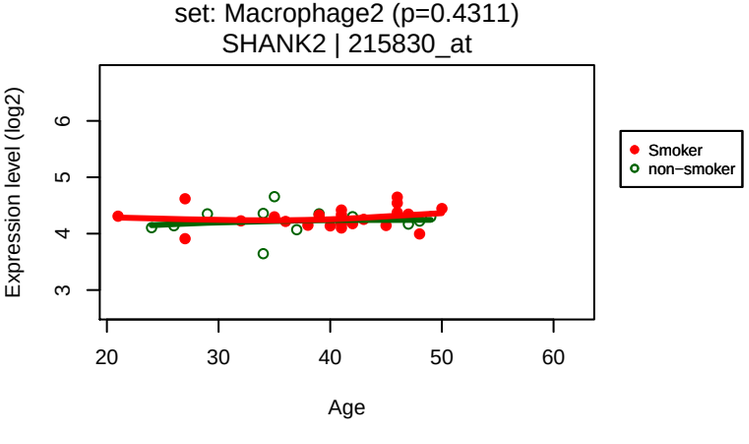

set: SmallAirway2 (p=0.4205)
SHANK2 | 215830_at
Expression level (log2)
Age
Smoker
non−smoker

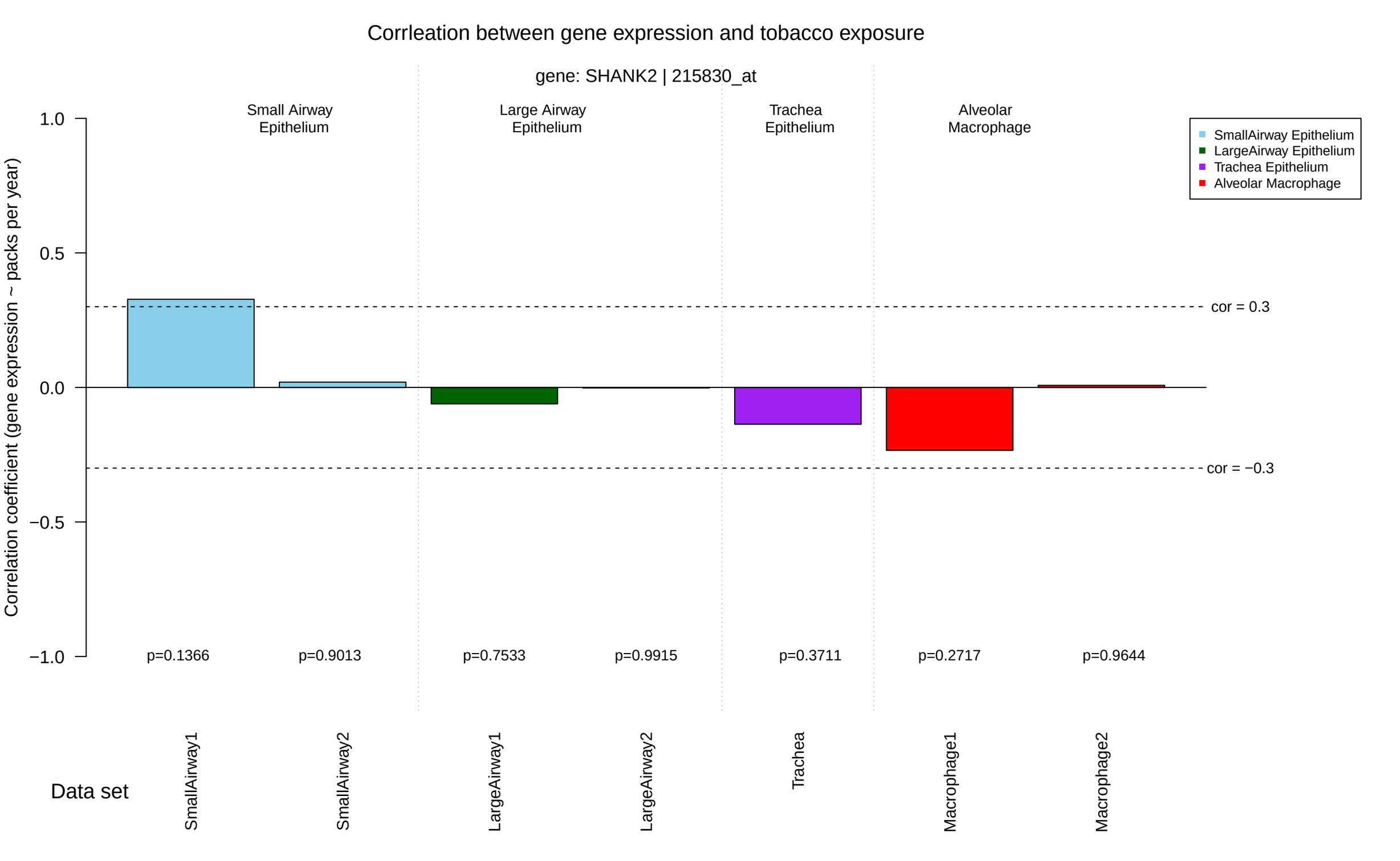

set: LargeAirway1 (p=0.6752)
SHANK2 | 215830_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3113)
SHANK2 | 215830_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2665)
SHANK2 | 215830_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7656)
SHANK2 | 215830_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4311)
SHANK2 | 215830_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SHANK2 | 215830_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1366
p=0.9013
p=0.7533
p=0.9915
p=0.3711
p=0.2717
p=0.9644
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2